

Theorem (Chebyshev's sum inequality). *Let*

$$a_1 \leq a_2 \leq \cdots \leq a_n, \quad b_1 \leq b_2 \leq \cdots \leq b_n$$

be real numbers. Then

$$n \sum_{i=1}^n a_i b_i \geq \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right), \quad (1)$$

with equality iff at least one of the two sequences is constant ($a_1 = a_2 = \cdots = a_n$ or $b_1 = b_2 = \cdots = b_n$).

Proof. 1. Degenerate cases. If $n = 1$ both sides of (??) coincide, so the inequality holds with equality. If all a_i are equal, say $a_i \equiv c$, then the left-hand side of (??) is $nc \sum b_i$ and the right-hand side is $c \sum b_i \cdot n$, which are identical; the same argument applies when all b_i are equal. Hence the statement is true in these degenerate situations. In the sequel we assume $n \geq 2$ and that neither sequence is constant.

2. Algebraic reformulation. Set

$$\bar{a} := \frac{1}{n} \sum_{i=1}^n a_i, \quad \bar{b} := \frac{1}{n} \sum_{i=1}^n b_i.$$

Then

$$\begin{aligned} n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) &= n \sum_{i=1}^n a_i b_i - n^2 \bar{a} \bar{b} \\ &= n \left[\sum_{i=1}^n a_i b_i - n \bar{a} \bar{b} \right]. \end{aligned}$$

Moreover

$$\sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b}) = \sum_{i=1}^n a_i b_i - n \bar{a} \bar{b}.$$

Consequently

$$n \sum_{i=1}^n a_i b_i - \left(\sum_{i=1}^n a_i \right) \left(\sum_{i=1}^n b_i \right) = n \sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b}). \quad (2)$$

Since $n > 0$, inequality (??) is equivalent to

$$\sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b}) \geq 0. \quad (3)$$

Define the centred sequences

$$\alpha_i := a_i - \bar{a}, \quad \beta_i := b_i - \bar{b} \quad (i = 1, \dots, n).$$

Subtracting a constant does not change the order, so

$$\alpha_1 \leq \alpha_2 \leq \cdots \leq \alpha_n, \quad \beta_1 \leq \beta_2 \leq \cdots \leq \beta_n.$$

Thus (α_i) and (β_i) are **similarly ordered**.

3. Application of the rearrangement inequality. The rearrangement inequality states that for any permutation σ of $\{1, \dots, n\}$,

$$\sum_{i=1}^n \alpha_i \beta_i \geq \frac{1}{n!} \sum_{\sigma} \sum_{i=1}^n \alpha_i \beta_{\sigma(i)}. \quad (4)$$

Because the centred sequences have zero sum,

$$\sum_{i=1}^n \alpha_i = 0 = \sum_{i=1}^n \beta_i,$$

the inner double sum equals 0, and therefore the right-hand side of (4) is 0. Hence

$$\sum_{i=1}^n \alpha_i \beta_i \geq 0.$$

Using the definitions of α_i, β_i this is precisely (1), and by (1) we obtain (2).

4. Equality condition. Equality in the rearrangement inequality occurs iff the two sequences are equal up to a permutation or one of them is constant. Since $\sum \alpha_i = 0 = \sum \beta_i$, the “equal up to a permutation” case forces every α_i and β_i to be zero; otherwise the sum $\sum \alpha_i \beta_i$ could not be zero. Hence equality in (2) (and thus in (1)) holds precisely when

- all $\alpha_i = 0$, i.e. $a_1 = a_2 = \cdots = a_n$, or
- all $\beta_i = 0$, i.e. $b_1 = b_2 = \cdots = b_n$.

These are exactly the cases stated in the theorem (the case $n = 1$ being included).

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